

Consider a clickstream aggregation job that groups events by user session identifier. During a flash sale, a small number of promotional sessions generate an outsized share of events, and any static hash-based partitioning scheme routes a disproportionate volume of traffic to whichever worker owns those hot keys. An adaptive scheme instead detects the emerging concentration of traffic and redistributes the hot keys before the imbalance manifests as elevated tail latency, without operator intervention or a job restart.